

Post-Graduate Common Admission Test (PG CAT)

Official Detailed Syllabus: M.Sc Mathematics (Test Code 611)

Program: M.Sc in
Mathematics

Level: Post-Graduate Entrance
Examination

Eligibility: B.Sc Mathematics
Degree

Parameter	Examination Specification
Total Questions	150 Objective Multiple Choice Questions
Duration	2 Hours (120 Minutes)
Marking Scheme	+3 marks for correct response; -1 mark for incorrect response
Core Focus	Undergraduate (B.Sc) Mathematics core curriculum: Analysis, Algebra, Topology, Calculus, ODE/PDE

1. Real Analysis

- Real number system as a complete ordered field, Archimedean property, supremum, infimum.
- Sequences and series: convergence, limsup, liminf, Bolzano-Weierstrass theorem, Cauchy criterion, tests of convergence.
- Functions of single variable: limits, continuity, uniform continuity, intermediate value property.
- Differentiability, Mean Value Theorems, Taylor's theorem with remainder, maxima and minima.
- Riemann integration: partition, upper/lower sums, integrability criteria, fundamental theorem of calculus.
- Sequences and series of functions: pointwise and uniform convergence, Weierstrass M-test.

2. Linear Algebra

- Vector spaces, subspaces, linear dependence, linear independence, basis and dimension, coordinates.
- Linear transformations, rank-nullity theorem, matrix representations of linear transformations, change of basis.
- Systems of linear equations, Gauss elimination, determinants and properties.
- Eigenvalues, eigenvectors, characteristic polynomial, Cayley-Hamilton theorem, minimal polynomial.
- Diagonalizability, invariant subspaces, inner product spaces, Gram-Schmidt orthogonalization process, self-adjoint and unitary operators.

3. Abstract Algebra

- Group theory: groups, subgroups, cyclic groups, permutation groups, symmetric and alternating groups.
- Cosets, Lagrange's theorem, normal subgroups, quotient groups, group homomorphisms, isomorphism theorems.
- Automorphisms, Cayley's theorem, class equation, Sylow's theorems overview.
- Ring theory: rings, integral domains, division rings, fields, ideals, prime and maximal ideals, quotient rings.
- Homomorphisms of rings, polynomial rings, divisibility in integral domains, Euclidean domains, principal ideal domains (PID), unique factorization domains (UFD).

4. Complex Analysis

- Algebra of complex numbers, geometric representation, analytic functions, Cauchy-Riemann equations, harmonic functions.
- Conformal mappings, Mobius (bilinear) transformations.
- Complex integration: contours, Cauchy-Goursat theorem, Cauchy's integral formula, Liouville's theorem, fundamental theorem of algebra.
- Taylor series, Laurent series, singularities, residues, Cauchy's residue theorem and evaluation of real integrals.

5. Differential Equations (ODE & PDE)

- First-order ordinary differential equations: exact, linear, Bernoulli, integrating factors.
- Higher-order linear differential equations with constant coefficients, method of undetermined coefficients, variation of parameters.
- Cauchy-Euler equations, system of linear differential equations.
- Partial differential equations: formation, first-order linear and quasilinear PDE, Lagrange's method, Charpit's method.
- Classification of second-order linear PDE: Laplace, wave, and heat equations in standard forms.

6. Metric Spaces & General Topology

- Metric spaces: open sets, closed sets, interior, closure, boundary, limit points.
- Completeness, Cantor intersection theorem, Baire category theorem.
- Continuous mappings between metric spaces, uniform continuity. Compactness, Heine-Borel theorem, connectedness and path-connectedness.
- Topological spaces: bases, subbases, subspace topology, product topology, Hausdorff spaces.

7. Numerical Analysis, Vector Calculus & Probability

- Numerical solutions of algebraic equations: bisection method, Newton-Raphson method, secant method.
- Interpolation: Lagrange interpolation, Newton's divided difference. Numerical integration: trapezoidal and Simpson's rules.
- Vector calculus: gradient, divergence, curl, line, surface and volume integrals, Green's, Stokes', and Gauss' divergence theorems.
- Probability: axioms of probability, conditional probability, Bayes theorem, random variables, expectation, variance, discrete and continuous distributions.